Practical - 11

Practical Title: Student Management CRUD with Search and Filter

Problem Definition: Create a student management module to add, view, edit, delete, search and filter student records. All database operations must use MYSQLi statements and clear success/failure messages.

A. Theory

1. Are SELECT, INSERT, UPDATE, and DELETE operations working?

Answer: Yes, full SELECT, INSERT, UPDATE and DELETE queries are implemented.

2. Is search by name/email/course implemented?

Answer: Yes, SQL LIKE conditions query records across student name, roll number, and email.

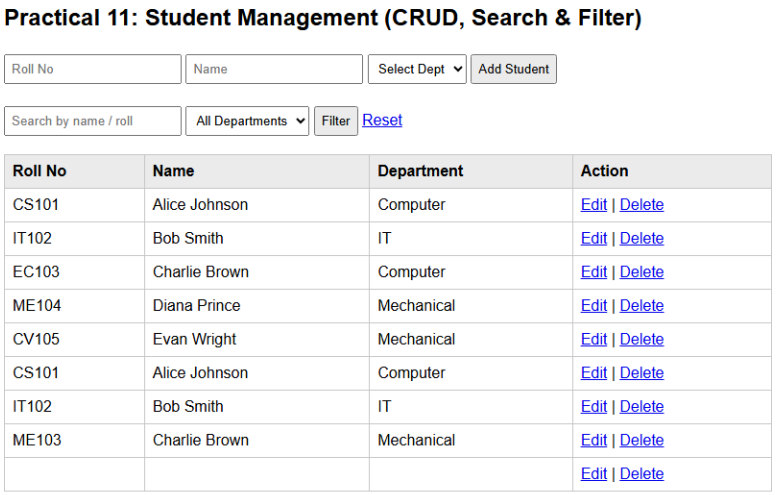
3. Are delete operations confirmed before execution?

Answer: Yes, JavaScript confirm() dialog prompts the user before sending the delete request.

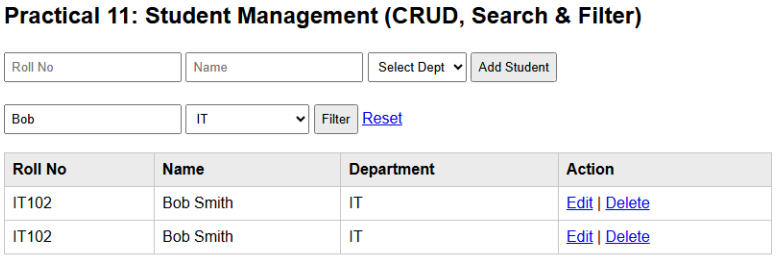
B. Implementation/ Output/Result

Feature	Implementation
Create	INSERT query using MySQLi prepared statements.
Read	SELECT query rendering records in a table.
Update	Pre-fills form and runs UPDATE query.
Delete	Runs DELETE query with JS confirm() prompt.
Search and filter	Dynamic SQL LIKE and WHERE department = ?

C. Evidence of Work (Screenshots)



Description: Figure 11.1 records view



Description: Figure 11.2 search and filter

D. GitHub Repository Link:

Link: https://github.com/prep04-yl/StudentHub-Portal

E. Challenges Faced:

Preventing SQL injection when combining multi-fields search conditions with dynamic filters.

F. Learning Outcome:

I learned how to develop a complete and secure data-mangement page.

Faculty Remarks: **(Keep this blank for faculty)**